NASA EXOPLANET ARCHIVE
NASA EXOPLANET SCIENCE INSTITUTE

Planetary Systems and Planetary Systems Composite Parameters Table Definitions

This page describes the parameters available in the Planetary Systems (`PS`) and Planetary Systems Composite Parameters (`PSCompPars`) interactive tables. These parameters can also be retrieved programmatically through the archive's Table Access Protocol (TAP) service.

Skip to a subsection:

Choosing the Right Table For Your Needs

Top

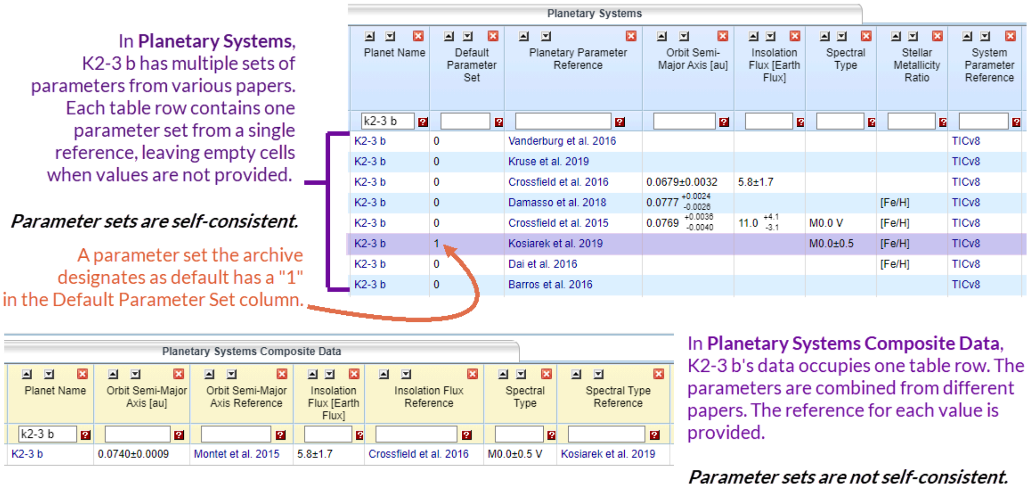
Choosing data from the PS or PSCompPars tables depends on your use case:

- **PS** provides a single table view of the all of the ingested planetary systems for each known exoplanet with each row containing a self-contained set of parameters (planet + stellar + system) for each reference. **The PS Table contains one row per planet per reference.**
- **PSCompPars** is a more filled-in table, with only one row per planet, enabling a more statistical view of the known exoplanet population and their host environments. **This table provides a more complete, though not necessarily self-consistent, set of parameters.**
    - For an explanation on how the archive calculates values for this table, read How the Archive Calculates Values in the Planetary Systems Composite Table

For a comparison between the PS and PSComPars tables, see the following graphic:

(*Click to enlarge*)

Comparing the Planetary Systems and Planetary Systems Composite Data Tables



Query-based Table Access

Top

The PS and PSCompPars tables replaced the Confirmed Planets, Composite Parameters, and Extended Planet Data tables that were retired in April 2021. (For details, see the Transition page.)

If you were using the archive's application programming interface (API) to retrieve data from the retired tables, you need to create a new query to work with the archive's TAP service that points to one of the new tables. Please note the database column names have also changed; refer to the following documents to identify the appropriate database names for your scripts/queries:

- Master list of columns in the PS, PSCompPars, Confirmed Planets, Extended Planet Data, and Composite Parameters tables: PDF CSV
- Column mapping between the new PS Table and the retired Confirmed Planets and Extended Planet Data tables: PDF CSV
- Column mapping between the new PSCompPars Table and the retired Composite Parameters Table: PDF CSV
- Parameters that were in the retired Confirmed Planets, Extended Planet Data and Composite Parameters tables, but NOT in the PS Table: PDF CSV

Data Column Definitions

Top

Additional Default Columns

The parameters in this section are displayed when the interactive table is first loaded, when **Reset Filters** is clicked, or when a TAP query when all default columns are retrieved, but do not fit in any of the other categories listed on this page. All default columns on this page are noted with a †.

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
default_flag†	Default Parameter Set	Boolean flag indicating whether given set of planet parameters has been selected as default (1=yes, 0=no)	X	
soltype†	Solution Type	Disposition of planet according to given planet parameter set	X	
pl_controv_flag†	Controversial Flag	Flag indicating whether the confirmation status of a planet has been questioned in the published literature (1=yes, 0=no)	X	X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Names

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table

pl_name†	Planet Name	Planet name most commonly used in the literature	X	X
hostname†	Host Name	Stellar name most commonly used in the literature	X	X
pl_letter	Planet Letter	Letter assigned to the planetary component of a planetary system	X	X
hd_name	HD ID	Name of the star as given by the Henry Draper Catalog	X	X
hip_name	HIP ID	Name of the star as given by the Hipparcos Catalog	X	X
tic_id	TIC ID	Name of the star as given by the TESS Input Catalog	X	X
gaia_dr2_id	Gaia DR2 ID	Name of the star as given by the Gaia DR2 Catalog	X	X
gaia_dr3_id	Gaia DR3 ID	Name of the star as given by the Gaia DR3 Catalog	X	X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

System Composition

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
sy_snum†	Number of Stars	Number of gravitationally bound stars in the planetary system	X	X
sy_pnum†	Number of Planets	Number of confirmed planets in the planetary system	X	X
sy_mnum	Number of Moons	Number of moons in the planetary system	X	X
cb_flag	Circumbinary Flag	Flag indicating whether the planet orbits a binary system (1=yes, 0=no)	X	X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Planet Discovery

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
discoverymethod†	Discovery Method	Method by which the planet was first identified	X	X
disc_year†	Discovery Year	Year the planet was discovered	X	X
disc_refname	Discovery Reference	Reference name for discovery publication	X	X
disc_pubdate	Discovery Publication Date	Publication Date of the planet discovery referee publication	X	X
disc_locale	Discovery Locale	Location of observation of planet discovery (Ground or Space)	X	X
disc_facility†	Discovery Facility	Name of facility of planet discovery observations	X	X
disc_telescope	Discovery Telescope	Name of telescope of planet discovery observations	X	X
disc_instrument	Discovery Instrument	Name of instrument of planet discovery observations	X	X

Top

Detections

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
rv_flag	Detections by Radial Velocity Variations	Flag indicating if the planet host star exhibits radial velocity variations due to the planet (1=yes, 0=no)	X	X
pul_flag	Detected by Pulsar Timing Variations	Boolean flag indicating if the planet host star exhibits pulsar timing variations due to the planet (1=yes, 0=no)	X	X
ptv_flag	Detected by Pulsation Timing Variations	Boolean flag indicating if the planet host star exhibits pulsation timing variations due to the planet (1=yes, 0=no)	X	X
tran_flag	Detected by Transits	Flag indicating if the planet transits its host star (1=yes, 0=no)	X	X
ast_flag	Detected by Astrometric Variations	Flag indicating if the planet host star exhibits astrometrical variations due to the planet (1=yes, 0=no)	X	X

obm_flag	Detected by Orbital Brightness Modulations	Flag indicating whether the planet exhibits orbital modulations on the phase curve (1=yes, 0=no)	X	X
micro_flag	Detected by Microlensing	Boolean flag indicating if the planetary system acted as a lens during an observed microlensing event (1=yes, 0=no)	X	X
etv_flag	Detected by Eclipse Timing Variations	Flag indicating whether a circumbinary planet that orbits an eclipsing binary induces eclipse timing variations (ETVs) in the binary pair (1=yes, 0=no)	X	X
ima_flag	Detected by Imaging	Flag indicating if the planet has been observed via imaging techniques (1=yes, 0=no)	X	X
dkin_flag	Detection by Disk Kinematics	Boolean flag indicating if the presence of the planet was inferred due to its kinematic influence on the protoplanetary disk of its host star (1=yes, 0=no)	X	X

Top

Planet Parameters

Database Column Name	Table Label	Description	Uncertainties Column (positive +) (negative -)	Limit Column	In PS Table	In PSCompPars Table
pl_refname†	Planetary Parameter Reference	Reference of publication used for given planet parameter set			X	
pl_orbper†	Orbital Period [days]	Time the planet takes to make a complete orbit around the host star or system	(+) pl_orbpererr1 (-) pl_orbpererr2	pl_orbperlim	X	X
pl_orbper_reflink	Orbital Period Reference	Reference of publication used for given planet parameter set				X
pl_orbsmax†	Orbit Semi-Major Axis [au]	The longest radius of an elliptic orbit, or, for exoplanets detected via gravitational microlensing or direct imaging, the projected separation in the plane of the sky	(+) pl_orbsmaxerr1 (-) pl_orbsmaxerr2	pl_orbsmaxlim	X	X
pl_orbsmax_reflink	Orbit Semi-Major Axis Reference	Reference of publication used for given parameter set				X
pl_angsep	Angular Separation [mas]	Angular separation on the sky between the star and planet. **Note:** This may be derived in several ways, but is typically either an actual reported angular separation at a given epoch (for which the literature source will be provided in the reference column), or a calculated angular separation using the semi-major axis and the distance to the system (which may or may not be equivalent to the maximum angular separation, depending on orbital eccentricity and orientation). Refer to original data source for further information.	(+) pl_angseperr1 (-) pl_angseperr2	pl_angseplim		X
pl_angsep_reflink	Angular Separation Reference	Reference of publication used for given parameter set				X

pl_rade†	Planet Radius [Earth Radius]	Length of a line segment from the center of the planet to its surface, measured in units of radius of the Earth	(+) pl_radeerr1 (-) pl_radeerr2	pl_radelim	X	X
pl_rade_reflink	Planet Radius [Earth Radius] Reference	Reference of publication used for given parameter set				X
pl_radj†	Planet Radius [Jupiter Radius]	Length of a line segment from the center of the planet to its surface, measured in units of radius of Jupiter	(+) pl_radjerr1 (-) pl_radjerr2	pl_radjlim	X	X
pl_radj_reflink	Planet Radius [Jupiter Radius] Reference	Reference of publication used for given parameter set				X
pl_masse	Planet Mass [Earth Mass]	Amount of matter contained in the planet, measured in units of masses of the Earth	(+) pl_masseerr1 (-) pl_masseerr2	pl_masselim	X	
pl_massj	Planet Mass [Jupiter Mass]	Amount of matter contained in the planet, measured in units of masses of Jupiter	(+) pl_massjerr1 (-) pl_massjerr2	pl_massjlim	X	
pl_msinie	Planet Mass*sin(i) [Earth Mass]	Minimum mass of a planet as measured by radial velocity, measured in units of masses of the Earth	(+) pl_msinieerr1 (-) pl_msinieerr2	pl_msinielim	X	
pl_msinij	Planet Mass*sin(i) [Jupiter Mass]	Minimum mass of a planet as measured by radial velocity, measured in units of masses of Jupiter	(+) pl_msinijerr1 (-) pl_msinijerr2	pl_msinijlim	X	
pl_cmasse	Planet Mass*sin(i)/sin(i) [Earth Mass]	A calculated quantity indicating the quotient of the lower limit of the measured planet mass, denoted as its mass times the sine of its inclination, and the sine of its inclination, measured in units of the mass of the Earth. This is specified for references in which the inclination is provided as well as the planet mass limit, but the true mass is not reported.	(+) pl_cmasseerr1 (-) pl_cmasseerr2	pl_cmasselim	X	
pl_cmassj	Planet Mass*sin(i)/sin(i) [Jupiter Mass]	A calculated quantity indicating the quotient of the lower limit of the measured planet mass, denoted as its mass times the sine of its inclination, and the sine of its inclination, measured in units of the mass of Jupiter. This is specified for references in which the inclination is provided as well as the planet mass limit, but the true mass is not reported.	(+) pl_cmassjerr1 (-) pl_cmassjerr2	pl_cmassjlim	X	
pl_bmasse†	Planet Mass or Mass*sin(i) [Earth Mass]	Best planet mass estimate available, in order of preference: Mass, M*sin(i)/sin(i), or M*sin(i), depending on availability, and measured in Earth masses	(+) pl_bmasseerr1 (-) pl_bmasseerr2	pl_bmasselim	X	X
pl_bmasse_reflink	Planet Mass or Mass*sin(i) [Earth Mass] Reference	Reference of publication used for given parameter set				X

pl_bmassj†	Planet Mass or Mass*sin(i) [Jupiter Mass]	Best planet mass estimate available, in order of preference: Mass, M*sin(i)/sin(i), or M*sin(i), depending on availability, and measured in Jupiter masses	(+) pl_bmassjerr1 (-) pl_bmassjerr2	pl_bmassjlim	X	X
pl_bmassj_reflink	Planet Mass or Mass*sin(i) [Jupiter Mass] Reference	Reference of publication used for given parameter set				X
pl_bmassprov†	Planet Mass or Mass*sin(i) Provenance	Provenance of the measurement of the best mass. Options are: Mass, M*sin(i)/sin(i), and M*sini			X	X
pl_dens	Planet Density [g/cm**3]	Amount of mass per unit of volume of the planet	(+) pl_denserr1 (-) pl_denserr2	pl_denslim	X	X
pl_dens_reflink	Planet Density [g/cm**3] Reference	Reference of publication used for given parameter set				
pl_orbeccen†	Eccentricity	Amount by which the orbit of the planet deviates from a perfect circle	(+) pl_orbeccenerr1 (-) pl_orbeccenerr2	pl_orbeccenlim	X	X
pl_orbeccen_reflink	Eccentricity Reference	Reference of publication used for given parameter set				
pl_insol†	Insolation Flux [Earth Flux]	Insolation flux is another way to give the equilibrium temperature. It's given in units relative to those measured for the Earth from the Sun.	(+) pl_insolerr1 (-) pl_insolerr2	pl_insollim	X	X
pl_insol_reflink	Insolation Flux [Earth Flux] Reference	Reference of publication used for given parameter set				X
pl_eqt†	Equilibrium Temperature [K]	The equilibrium temperature of the planet as modeled by a black body heated only by its host star, or for directly imaged planets, the effective temperature of the planet required to match the measured luminosity if the planet were a black body	(+) pl_eqterr1 (-) pl_eqterr2	pl_eqtlim	X	X
pl_eqt_reflink	Equilibrium Temperature [K] Reference	Reference of publication used for given parameter set				X
pl_orbincl	Inclination [deg]	Angle of the plane of the orbit relative to the plane perpendicular to the line-of-sight from Earth to the object	(+) pl_orbinclerr1 (-) pl_orbinclerr2	pl_orbincllim	X	X
pl_tranmid	Time of Conjunction (Transit Midpoint) [days]	The time given by the average of the time the planet begins to cross the stellar limb and the time the planet finishes crossing the stellar limb.	(+) pl_tranmiderr1 (-) pl_tranmiderr2	pl_tranmidlim	X	X
pl_tranmid_systemref	Transit Midpoint Time Reference Frame and Standard	Time system basis for time of conjunction (Transit Midpoint)				X
pl_tranmid_reflink	Time of Conjunction (Transit Midpoint) Reference	Reference of publication used for given parameter set				X

pl_tsystemref	Time Reference Frame and Standard	Time system basis for temporal and orbital parameters			X	
ttv_flag†	Data show Transit Timing Variations	Flag indicating if the planet orbit exhibits transit timing variations from another planet in the system (1=yes, 0=no). **Note:** Non-transiting planets discovered via the transit timing variations of another planet in the system will not have their TTV flag set, since they do not themselves demonstrate TTVs.			X	X
pl_imppar	Impact Parameter	The sky-projected distance between the center of the stellar disc and the center of the planet disc at conjunction, normalized by the stellar radius	(+) pl_impparerr1 (-) pl_impparerr2	pl_impparlim	X	X
pl_imppar_reflink	Impact Parameter Reference	Reference of publication used for given parameter set				X
pl_trandep	Transit Depth [%]	The size of the relative flux decrement caused by the orbiting body transiting in front of the star	(+) pl_trandeperr1 (-) pl_trandeperr2	pl_trandeplim	X	X
pl_trandep_reflink	Transit Depth [%] Reference	Reference of publication used for given parameter set				X
pl_trandur	Transit Duration [hrs]	The length of time from the moment the planet begins to cross the stellar limb to the moment the planet finishes crossing the stellar limb	(+) pl_trandurerr1 (-) pl_trandurerr2	pl_trandurlim	X	X
pl_trandur_reflink	Transit Duration [hrs] Reference	Reference of publication used for given parameter set				X
pl_ratdor	Ratio of Semi-Major Axis to Stellar Radius	The distance between the planet and the star at mid-transit divided by the stellar radius. For the case of zero orbital eccentricity, the distance at mid-transit is the semi-major axis of the planetary orbit.	(+) pl_ratdorerr1 (-) pl_ratdorerr2	pl_ratdorlim	X	X
pl_ratdor_reflink	Ratio of Semi-Major Axis to Stellar Radius Reference	Reference of publication used for given parameter set				X
pl_ratror	Ratio of Planet to Stellar Radius	The planet radius divided by the stellar radius	(+) pl_ratrorerr1 (-) pl_ratrorerr2	pl_ratrorlim	X	X
pl_ratror_reflink	Ratio of Planet to Stellar Radius Reference	Reference of publication used for given parameter set				X
pl_occdep	Occultation Depth [%]	Depth of occultation of secondary eclipse	(+) pl_occdeperr1 (-) pl_occdeperr2	pl_occdeplim	X	X
pl_occdep_reflink	Occultation Depth [%] Reference	Reference of publication used for given parameter set				X
pl_orbtper	Epoch of Periastron [deg]	The time of the planet's periastron passage	(+) pl_orbtpererr1 (-) pl_orbtpererr2	pl_orbtperlim	X	X
pl_orbtper_reflink	Epoch of Periastron [deg] Reference	Reference of publication used for given parameter set				X

pl_orblper	Argument of Periastron [deg]	The angular separation between the orbit's ascending node and periastron. Note: there are a varying conventions in the exoplanet literature regarding argument of periastron (or periapsis). For example, some publications refer the planet's orbit, others to the host star's reflex orbit, which differs by 180 deg. The values in the Exoplanet Archive are not corrected to a standardized system, but are as-reported for each publication.	(+) pl_orblpererr1 (-) pl_orblpererr2	pl_orblperlim	X	X
pl_rvamp	Radial Velocity Amplitude [m/s]	Half the peak-to-peak amplitude of variability in the stellar radial velocity	(+) pl_rvamperr1 (-) pl_rvamperr2	pl_rvamplim	X	X
pl_rvamp_reflink	Radial Velocity Amplitude Reference [m/s]	Reference of publication used for given parameter set				X
pl_projobliq	Projected Obliquity [deg]	The angle between the angular momentum vector of the rotation of the host star and the angular momentum vector of the orbit of the planet, projected into the plane of the sky. Depending on the choice of coordinate system, projected obliquity is represented in the literature as either lambda (λ) or beta (β), where λ is defined as the negative of β (i.e., λ = -β). Since λ is reported more often than β, all values of β have been converted to λ.	(+) pl_projobliqerr1 (-) pl_projobliqerr2	pl_projobliqlim	X	X
pl_projobliq_reflink	Projected Obliquity [deg] Reference	Reference of publication used for given parameter set				X
pl_trueobliq	True Obliquity [deg]	The angle between the angular momentum vector of the rotation of the host star and the angular momentum vector of the orbit of the planet	(+) pl_trueobliqerr1 (-) pl_trueobliqerr2	pl_trueobliqlim	X	X
pl_trueobliq_reflink	True Obliquity [deg] Reference	Reference of publication used for given parameter set				X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Stellar Data

Database Column Name	Table Label	Description	Uncertainties Column (positive +) (negative -)	Limit Column	In PS Table	In PSCompPars Table
st_refname†	Stellar Parameter Reference	Reference of publication used for given stellar parameter set			X	
st_spectype	Spectral Type	Classification of the star based on their spectral			X	X

		characteristics following the Morgan-Keenan system				
st_spectype_reflink	Spectral Type Reference	Reference of publication used for given parameter set				X
st_teff†	Stellar Effective Temperature [K]	Temperature of the star as modeled by a black body emitting the same total amount of electromagnetic radiation	(+) st_tefferr1 (-) st_tefferr2	st_tefflim	X	X
st_teff_reflink	Stellar Effective Temperature [K] Reference	Reference of publication used for given parameter set				X
st_rad†	Stellar Radius [Solar Radius]	Length of a line segment from the center of the star to its surface, measured in units of radius of the Sun	(+) st_raderr1 (-) st_raderr2	st_radlim	X	X
st_rad_reflink	Stellar Radius [Solar Radius] Reference	Reference of publication used for given parameter set				X
st_mass†	Stellar Mass [Solar mass]	Amount of matter contained in the star, measured in units of masses of the Sun	(+) st_masserr1 (-) st_masserr2	st_masslim	X	X
st_mass_reflink	Stellar Mass [Solar mass] Reference	Reference of publication used for given parameter set				X
st_met†	Stellar Metallicity [dex]	Measurement of the metal content of the photosphere of the star as compared to the hydrogen content	(+) st_meterr1 (-) st_meterr2	st_metlim	X	X
st_met_reflink	Stellar Metallicity [dex] Reference	Reference of publication used for given parameter set				X
st_metratio†	Stellar Metallicity Ratio	Ratio for the Metallicity Value ([Fe/H] denotes iron abundance, [M/H] refers to a general metal content)	(+) st_metratioerr1 (-) st_metratioerr2	st_metratiolim	X	X
st_lum	Stellar Luminosity [log10(Solar)]	Amount of energy emitted by a star per unit time, measured in units of solar luminosities	(+) st_lumerr1 (-) st_lumerr2	st_lumlim	X	X
st_lum_reflink	Stellar Luminosity [log10(Solar)] Reference	Reference of publication used for given parameter set				X
st_logg†	Stellar Surface Gravity [log10(cm/s**2)]	Gravitational acceleration experienced at the stellar surface	(+) st_loggerr1 (-) st_loggerr2	st_logglim	X	X
st_logg_reflink	Stellar Surface Gravity [log10(cm/s**2)] Reference	Reference of publication used for given parameter set				X
st_age	Stellar Age [Gyr]	The age of the host star	(+) st_ageerr1 (-) st_ageerr2	st_agelim	X	X
st_age_reflink	Stellar Age [Gyr] Reference	Reference of publication used for given parameter set				X
st_dens	Stellar Density [g/cm**3]	Amount of mass per unit of volume of the star	(+) st_denserr1 (-) st_denserr2	st_denslim	X	X
st_dens_reflink	Stellar Density [g/cm**3] Reference	Reference of publication used for given parameter set				X

st_vsin	Stellar Rotational Velocity [km/s]	Rotational velocity at the equator of the star multiplied by the sine of the inclination	(+) st_vsinerr1 (-) st_vsinerr2	st_vsinlim	X	X
st_vsin_reflink	Stellar Rotational Velocity [km/s]	Reference of publication used for given parameter set				X
st_rotp	Stellar Rotational Period [days]	The time required for the planet host star to complete one rotation, assuming it is a solid body	(+) st_rotperr1 (-) st_rotperr2	st_rotplim	X	X
st_rotp_reflink	Stellar Rotational Period [days] Reference	Reference of publication used for given parameter set				X
st_radv	Systemic Radial Velocity [km/s]	Velocity of the star in the direction of the line of sight	(+) st_radverr1 (-) st_radverr2	st_radvlim	X	X
st_radv_reflink	Systemic Radial Velocity [km/s] Reference	Reference of publication used for given parameter set				X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

System Data

Database Column Name	Table Label	Description	Uncertainties Column (positive +) (negative -)	In PS Table	In PSCompPars Table
sy_refname†	System Parameter Reference	Reference of publication used for given system parameter set		X	
sy_pm	Total Proper Motion [mas/yr]	Angular change in position over time as seen from the center of mass of the Solar System	(+) sy_pmerr1 (-) sy_pmerr2	X	X
sy_pm_reflink	Total Proper Motion [mas/yr] Reference	Reference of publication used for given parameter set			X
sy_pmra	Proper Motion (RA) [mas/yr]	Angular change in right ascension over time as seen from the center of mass of the Solar System	(+) sy_pmraerr1 (-) sy_pmraerr2	X	X
sy_pmra_reflink	Proper Motion (RA) [mas/yr] Reference	Reference of publication used for given parameter set			X
sy_pmdec	Proper Motion (Dec) [mas/yr]	Angular change in declination over time as seen from the center of mass of the Solar System	(+) sy_pmdecerr1 (-) sy_pmdecerr2	X	X
sy_pmdec_reflink	Proper Motion (Dec) [mas/yr]	Reference of publication used for given parameter set			X
sy_dist†	Distance [pc]	Distance to the planetary system in units of parsecs	(+) sy_disterr1 (-) sy_disterr2	X	X
sy_dist_reflink	Distance [pc] Reference	Reference of publication used for given parameter set			X
sy_plx	Parallax [mas]	Difference in the angular position of a star as measured at two opposite positions within the Earth's orbit	(+) sy_plxerr1 (-) sy_plxerr2	X	X
sy_plx_reflink	Parallax [mas] Reference	Reference of publication used for given parameter set			X
pl_nnotes	Number of Notes	Number of Notes associated with the planet. View all notes in the Confirmed Planet Overview page.		X	X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Position (System Data Subset)

Database Column Name	Table Label	Description	Uncertainties Column (positive +) (negative -)	In PS Table	In PSCompPars Table
rastr†	RA [sexagesimal]	Right Ascension of the planetary system in sexagesimal format		X	X
decstr†	Dec [sexagesimal]	Declination of the planetary system in sexagesimal notation		X	X
ra	RA [decimal]	Right Ascension of the planetary system in decimal degrees	(+) raerr1 (-) raerr2	X	X
ra_reflink	RA [decimal] Reference	Reference of publication used for given parameter set			
dec	Dec [decimal]	Declination of the planetary system in decimal degrees	(+) decerr1 (-) decerr2	X	X
glat	Galactic Latitude [deg]	Galactic latitude of the planetary system in units of decimal degrees	(+) glaterr1 (-) glaterr2	X	X
glon	Galactic Longitude [deg]	Galactic longitude of the planetary system in units of decimal degrees	(+) glonerr1 (-) glonerr2	X	X
elat	Ecliptic Latitude [deg]	Ecliptic latitude of the planetary system in units of decimal degrees	(+) elaterr1 (-) elaterr2	X	X
elon	Ecliptic Longitude [deg]	Ecliptic longitude of the planetary system in units of decimal degrees	(+) elonerr1 (-) elonerr2	X	X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Photometry (System Data Subset)

Database Column Name	Table Label	Description	Uncertainties Column (positive +) (negative -)	In PS Table	In PSCompPars Table
sy_bmag	B (Johnson) Magnitude	Brightness of the host star as measured using the B (Johnson) band in units of magnitudes	(+) sy_bmagerr1 (-) sy_bmagerr2	X	X
sy_bymag_reflink	B (Johnson) Magnitude	Reference of publication used for given parameter set			X
sy_vmag†	V (Johnson) Magnitude	Brightness of the host star as measured using the V (Johnson) band in units of magnitudes	(+) sy_vmagerr1 (-) sy_vmagerr2	X	X
sy_vmag_reflink	V (Johnson) Magnitude	Reference of publication used for given parameter set			X
sy_jmag	J (2MASS) Magnitude	Brightness of the host star as measured using the J (2MASS) band in units of magnitudes	(+) sy_jmagerr1 (-) sy_jmagerr2	X	X
sy_jmag_reflink	J (2MASS) Magnitude Reference	Reference of publication used for given parameter set			X
sy_hmag	H (2MASS) Magnitude	Brightness of the host star as measured using the H (2MASS) band in units of magnitudes	(+) sy_hmagerr1 (-) sy_hmagerr2	X	X
sy_hmag_reflink	H (2MASS) Magnitude Reference	Reference of publication used for given parameter set			X
sy_kmag†	Ks (2MASS) Magnitude	Brightness of the host star as measured using the K (2MASS) band in units of magnitudes	(+) sy_kmagerr1 (-) sy_kmagerr2	X	X
sy_kmag_reflink	Ks (2MASS) Magnitude Reference	Reference of publication used for given parameter set			X
sy_umag	u (Sloan) Magnitude	Brightness of the host star as measured using the Sloan Digital Sky Survey (SDSS) u band, in units of magnitudes	(+) sy_umagerr1 (-) sy_umagerr2	X	X
sy_umag_reflink	u (Sloan) Magnitude Reference	Reference of publication used for given parameter set			X
sy_gmag	g (Sloan) Magnitude	Brightness of the host star as measured using the Sloan Digital Sky Survey (SDSS) g band, in	(+) sy_gmagerr1 (-) sy_gmagerr2	X	X

		units of magnitudes			
sy_gmag_reflink	g (Sloan) Magnitude	Reference of publication used for given parameter set			X
sy_rmag	r (Sloan) Magnitude	Brightness of the host star as measured using the Sloan Digital Sky Survey (SDSS) r band, in units of magnitudes	(+) sy_rmagerr1 (-) sy_rmagerr2	X	X
sy_rmag_reflink	r (Sloan) Magnitude	Reference of publication used for given parameter set			X
sy_imag	i (Sloan) Magnitude	Brightness of the host star as measured using the Sloan Digital Sky Survey (SDSS) i band, in units of magnitudes	(+) sy_imagerr1 (-) sy_imagerr2	X	X
sy_imag_reflink	i (Sloan) Magnitude	Reference of publication used for given parameter set			X
sy_zmag	z (Sloan) Magnitude	Brightness of the host star as measured using the Sloan Digital Sky Survey (SDSS) z band, in units of magnitudes	(+) sy_zmagerr1 (-) sy_zmagerr2	X	X
sy_zmag_reflink	z (Sloan) Magnitude Reference	Reference of publication used for given parameter set			X
sy_w1mag	W1 (WISE) Magnitude	Brightness of the host star as measured using the 3.4um (WISE) band in units of magnitudes.	(+) sy_w1magerr1 (-) sy_w1magerr2	X	X
sy_w1mag_reflink	W1 (WISE) Magnitude Reference	Reference of publication used for given parameter set			X
sy_w2mag	W2 (WISE) Magnitude	Brightness of the host star as measured using the 4.6um (WISE) band in units of magnitudes.	(+) sy_w2magerr1 (-) sy_w2magerr2	X	X
sy_w2mag_reflink	W2 (WISE) Magnitude Reference	Reference of publication used for given parameter set			X
sy_w3mag	W3 (WISE) Magnitude	Brightness of the host star as measured using the 12.um (WISE) band in units of magnitudes	(+) sy_w3magerr1 (-) sy_w3magerr2	X	X
sy_w3mag_reflink	W3 (WISE) Magnitude Reference	Reference of publication used for given parameter set			X
sy_w4mag	W4 (WISE) Magnitude	Brightness of the host star as measured using the 22.um (WISE) band in units of magnitudes	(+) sy_w4magerr1 (-) sy_w4magerr2	X	X
sy_w4mag_reflink	W4 (WISE) Magnitude Reference	Reference of publication used for given parameter set			X
sy_gaiamag†	Gaia Magnitude	Brightness of the host star as measuring using the Gaia band in units of magnitudes. Objects matched to Gaia using the Hipparcos or 2MASS IDs provided in Gaia DR2.	(+) sy_gaiamagerr1 (-) sy_gaiamagerr2	X	X
sy_gaiamag_reflink	Gaia Magnitude Reference	Reference of publication used for given parameter set			X
sy_icmag	I (Cousins) Magnitude	Brightness of the host star as measured using the I (Cousins) band in units of magnitudes.	(+) sy_icmagerr1 (-) sy_icmagerr2	X	X
sy_icmag_reflink	I (Cousins) Magnitude Reference	Reference of publication used for given parameter set			X
sy_tmag	TESS Magnitude	Brightness of the host star as measured using the TESS bandpass, in units of magnitudes	(+) sy_tmagerr1 (-) sy_tmagerr2	X	X
sy_tmag_reflink	TESS Magnitude Reference	Reference of publication used for given parameter set			X
sy_kepmag	Kepler Magnitude	Brightness of the host star as measured using the Kepler bandpass, in units of magnitudes	(+) sy_kepmagerr1 (-) sy_kepmagerr2	X	X

sy_kepmag_reflink	Kepler Magnitude Reference	Reference of publication used for given parameter set			X

† **Default column:** These columns display in the interactive table when the table is first loaded, and when **Reset Filters** is clicked.

Top

Dates (System Data Subset)

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
rowupdate†	Date of Last Update	Date of last update of the planet parameters	X	
pl_pubdate†	Planetary Parameter Reference Publication	Date of the publication of the given planet parameter set	X	
releasedate†	Release Date	Date that the given planet parameter set was publicly released by the NASA Exoplanet Archive	X	

Top

Additional Data

Database Column Name	Table Label	Description	In PS Table	In PSCompPars Table
st_nphot	Number of Photometry Time Series	Number of photometric time series records, including planet transit light curves, general transit light curves, and amateur light curves.	X	X
st_nrvc	Number of Radial Velocity Time Series	Number of literature radial velocity curves available for this star in the NASA Exoplanet Archive.	X	X
pl_ntranspec	Number of Transmission Spectroscopy Measurements	Number of literature transmission spectrum measurements for this planet in the NASA Exoplanet Archive	X	X
pl_nespec	Number of Eclipse Spectroscopy Measurements	Number of literature eclipse spectrum measurements for this planet in the NASA Exoplanet Archive	X	X
st_nspec	Number of Stellar Spectra Measurements	Number of literature spectra available for this star in the NASA Exoplanet Archive	X	X
pl_ndispec	Number of Direct Imaging Spectroscopy Measurements	Number of literature direct imaging spectrum measurements for this planet in the NASA Exoplanet Archive	X	X

Last updated: 14 July 2025

Home About Us Data Tools Support